

UCT41042 – Practical for Introduction to Smart System
Department of Information & Communication Technology
Faculty of Technology
South Eastern University of Sri Lanka
Lab Sheet No: 04

Date: 11.05.2026

Title: Interfacing Input & Output Devices with ESP32 in Wokwi

Aims: Learn how to correctly connect common Input Devices and Output Devices to the ESP32 Devkit in the Wokwi online simulator. This lab sheet focuses only on hardware connections (wiring) and includes a brief explanation of each device's purpose.

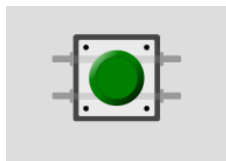
Tasks:

- Open wokwi and start a new project with ESP32 DevKit.
- Use the + button to search and add each component.
- Use different colored wires (Red = Power, Black = GND, others for signals).
- ESP32 GPIO pins are 3.3V logic. Always connect GND properly..

Input Devices

1. Pushbutton (Tactile Switch)

A momentary switch used to detect manual user input (press). Commonly used for triggering actions, resetting systems, or simple user interfaces.



Pins of device	Pins of ESP32
One leg (1.r)	GND
Other leg (2.r)	GPIO 5

2. Slide Switch / Toggle Switch

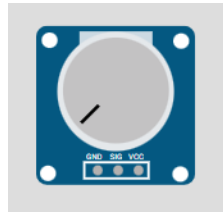
Maintains ON/OFF state for mode selection, power control, or configuration settings.



Pins of device	Pins of ESP32
Middle Pin (2)	GPIO 5
One outer Pin (1)	GND
Other outer pin (3)	-

3. Potentiometer (Rotary)

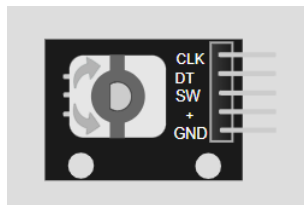
Variable resistor used to adjust analog values (e.g., volume, brightness, speed control).



Pins of device	Pins of ESP32
GND	GND
SIG	GPIO 34
VCC	3.3V

4. Rotary Encoder (KY-040)

Detects rotation direction and steps. Used for menu navigation, volume control, and precise parameter adjustment.



Pins of device	Pins of ESP32
CLK	GPIO 18
DT	GPIO 19
SW	GPIO 5
VCC	3.3V
GND	GND

5. Analog Joystick

Provides analog X & Y axis movement and a digital button. Used in gaming, robotics control, and pan-tilt systems.



Pins of device	Pins of ESP32
VCC	3.3V
VERT	GPIO 35
HORZ	GPIO 34
SEL	GPIO 5
GND	GND

6. Keypad

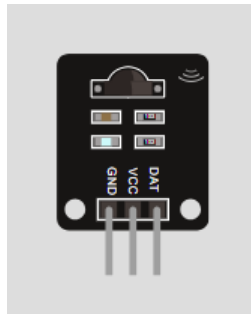
Allows numeric and character input. Used in password systems, calculators, and menu-based projects.



Pins of device	Pins of ESP32
R1	GPIO 19
R2	GPIO 18
R3	GPIO 5
R4	GPIO 17
C1	GPIO 16
C2	GPIO 4
C3	GPIO 2
C4	GPIO 15

7. IR Receiver

Detects and receives infrared signals from an IR Remote. Used for wireless remote control of devices (TVs, robots, home automation, etc.).



Pins of device	Pins of ESP32
GND	GND
VCC	3.3V
DAT	GPIO 4

Output Devices

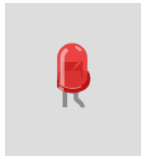
1. IR Remote

It acts like a TV remote. It transmits (sends) invisible infrared light signals when you press its buttons.



2. LED

Visual indicator for status, alerts, or debugging.



Pins of device	Pins of ESP32
C	GND
A	GPIO 5 (via 220Ω)

3. RGB LED

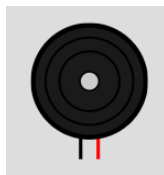
Produces multiple colors for status indication or decorative lighting.



Pins of device	Pins of ESP32
R	GPIO 5 (via 220Ω)
COM	GND
G	GPIO 18 (via 220Ω)
B	GPIO 19 (via 220Ω)

4. Buzzer

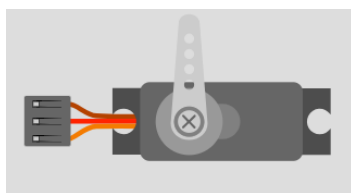
Produces simple beeps or tones for alerts, notifications, or feedback.



Pins of device	Pins of ESP32
1	GND
2	GPIO 5

5. Servo Motor

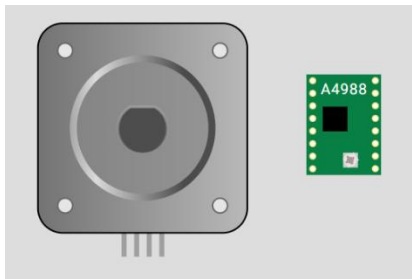
Precise angular control (0–180°). Used in robotics, camera gimbals, and door locks.



Pins of device	Pins of ESP32
GND	GND
V+	5V
PWM	GPIO 18

6. **Bipolar Stepper Motor + A4988 Driver**

Provides precise angular movement and position control. Used in robotics, 3D printers, CNC machines, and automated positioning systems.



Pins of Stepper motor	Pins of A4988 driver	Pins of ESP32
A-	2A	
A+	1A	
B+	1B	
B-	2B	
	RESET	Connect Together
	SLEEP	
	STEP	GPIO 32
	DIR	GPIO 19
	VDD	5V
	GND	GND

7. **Relay Module**

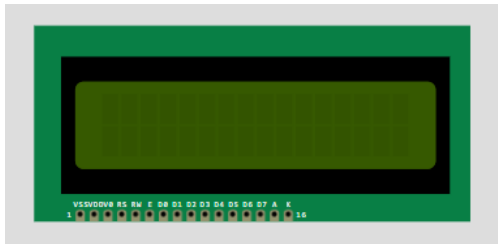
Allows low-power ESP32 to control high-power devices (lights, motors, appliances).



	Pins of device	Pins of ESP32
Input	VCC	5V
	GND	GND
	IN	GPIO 5
Output	NC	-
	COM	GND PIN of Output device
	NO	GND

8. LCD 16x2 (Parallel)

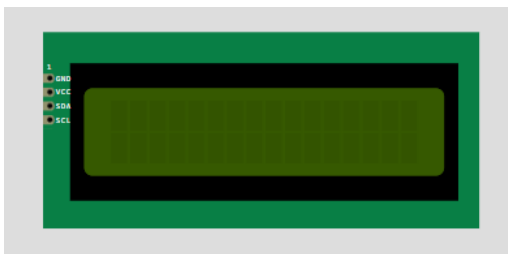
Displays text, numbers, and simple messages (2 rows × 16 characters). Classic and very common in beginner projects.



Pins of device	Pins of ESP32
VSS	GND
VDD	5V
V0	GND (via resistor)
RS	GPIO 19
RW	GND
E	GPIO 18
D0	-
D1	-
D2	-
D3	-
D4	GPIO 17
D5	GPIO 16
D6	GPIO 4
D7	GPIO 2
A	5V (via 220Ω)
K	GND

9. LCD 16x2 (I2C)

Same as above but uses only 2 wires (easier wiring).



Pins of device	Pins of ESP32
GND	GND
VCC	5V
SDA	GPIO 22
SCL	GPIO 21

10. MAX7219 LED Dot Matrix

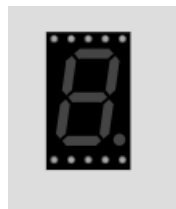
Displays scrolling text, animations, and patterns (8x8 matrix). Can daisy-chain multiple modules.



Pins of device	Pins of ESP32
V+	5V
GND	GND
DIN	GPIO 23
CS	GPIO 5
CLK	GPIO 18
V+.2	-
GND.2	-
DOUT	-
CS.2	-
CLK.2	-

11. Seven Segment Display

Displays numbers and limited characters. Widely used in clocks, counters, and scoreboards.



Pins of device	Pins of ESP32
A	GPIO 13 (via 220 Ω)
B	GPIO 12 (via 220 Ω)
C	GPIO 14 (via 220 Ω)
D	GPIO 27 (via 220 Ω)
E	GPIO 26(via 220 Ω)
F	GPIO 25 (via 220 Ω)
G	GPIO 33 (via 220 Ω)
DP	GPIO 32 (via 220 Ω)
COM.1	5V
COM.2	-